LAB EXPERIMENTS

Q3) Write a C program for Playfair algorithm is based on the use of a 5 X 5 matrix of letters constructed using a keyword. Plaintext is encrypted two letters at a time using this matrix.

Aim

To write a C program to implement the **Playfair Cipher** algorithm using a **5 × 5 matrix** generated from a keyword, and encrypt the plaintext two letters at a time.

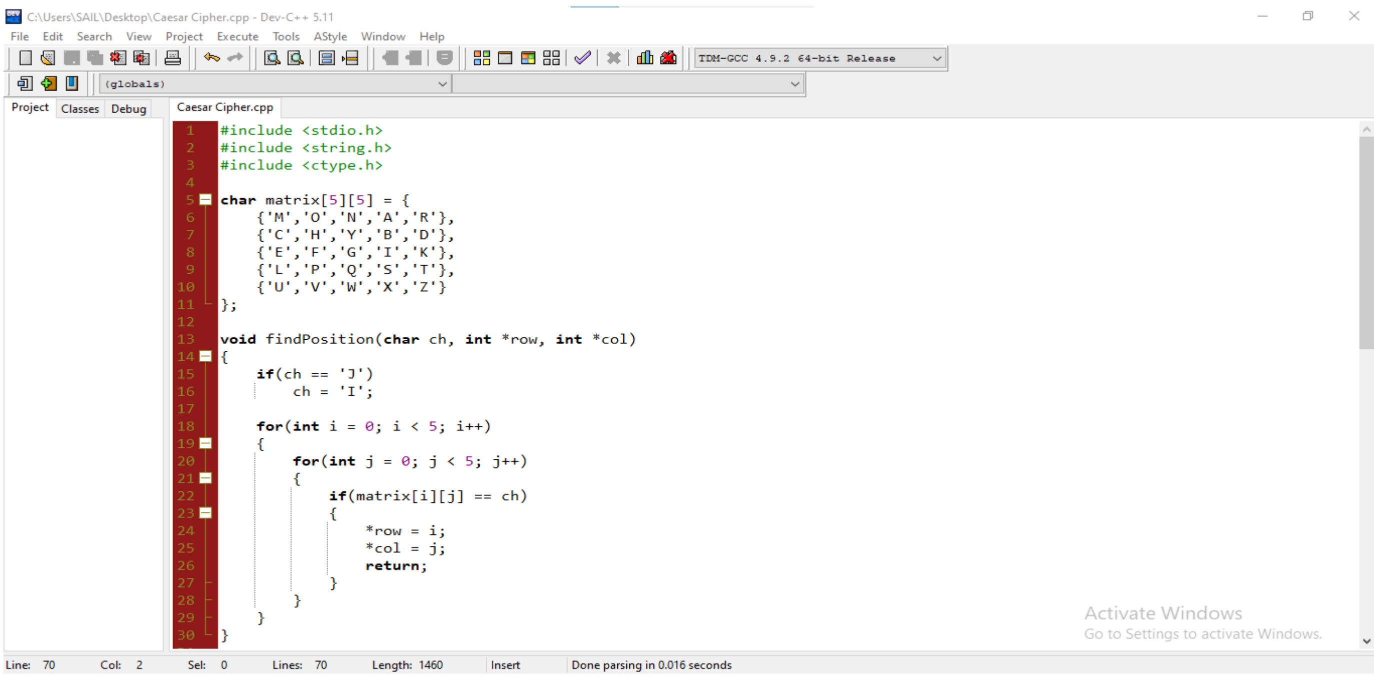
Algorithm

1. Start the program.
2. Enter the keyword and construct a **5 × 5 Playfair matrix** by removing duplicate letters and treating **I** and **J** as the same letter.
3. Fill the remaining positions in the matrix with the unused letters of the alphabet.
4. Read the plaintext from the user and convert it to uppercase.
5. Remove spaces, replace **J** with **I**, and divide the plaintext into pairs (digraphs). Insert **X** between repeated letters and append **X** if the length is odd.
6. For each pair of letters, locate their positions in the Playfair matrix.
7. Encrypt each pair using the Playfair rules:
8. If both letters are in the **same row**, replace each with the letter to its right.
9. If both letters are in the **same column**, replace each with the letter below it.
10. Otherwise, replace each letter with the one in the same row but in the column of the other letter.
11. Display the encrypted ciphertext.
12. Stop the program.

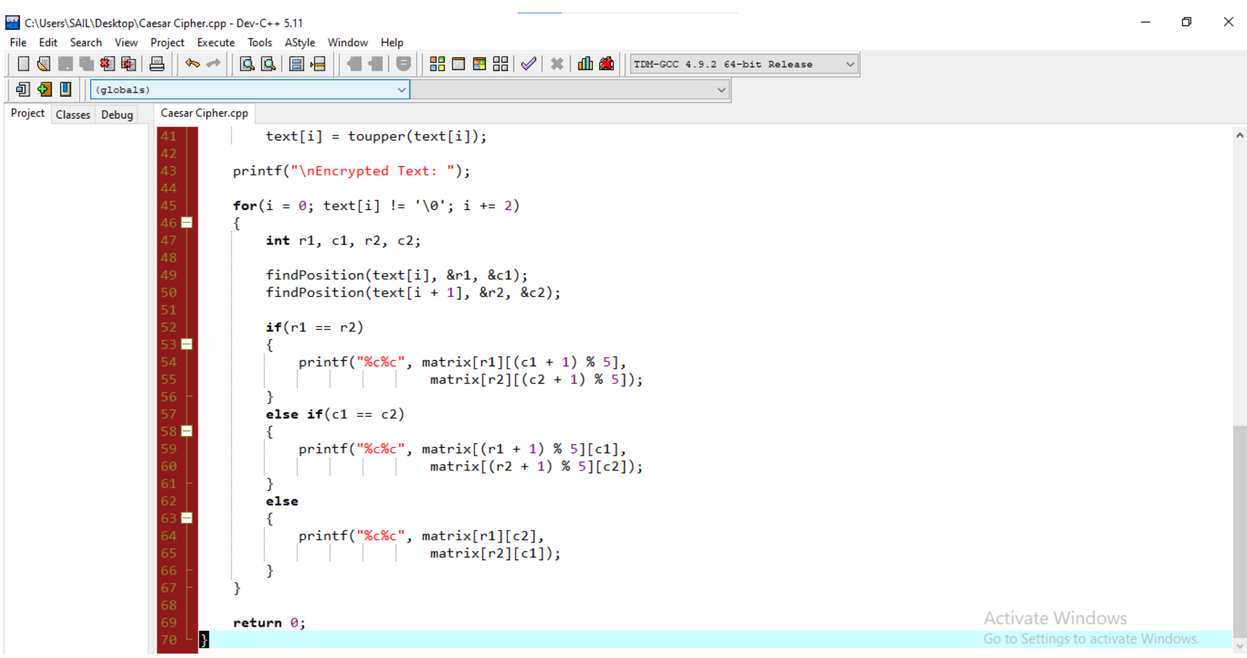
Result

Thus, the C program to implement the **Playfair Cipher** algorithm using a **5 × 5 matrix** was successfully executed, and the corresponding encrypted ciphertext was obtained.

Code:

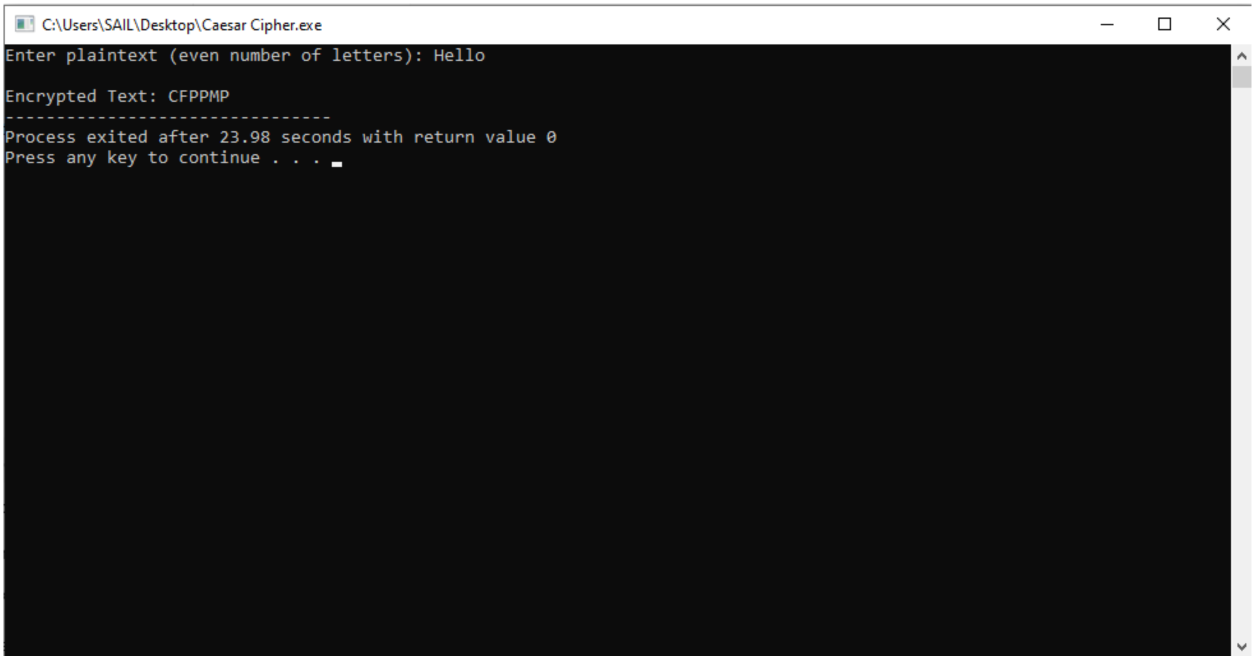


```c
#include <stdio.h>
#include <string.h>
#include <ctype.h>

char matrix[5][5] = {
    {'M','O','N','A','R'},
    {'C','H','Y','B','D'},
    {'E','F','G','I','K'},
    {'L','P','Q','S','T'},
    {'U','V','W','X','Z'}
};

void findPosition(char ch, int *row, int *col)
{
    if(ch == 'J')
        ch = 'I';

    for(int i = 0; i < 5; i++)
    {
        for(int j = 0; j < 5; j++)
        {
            if(matrix[i][j] == ch)
            {
                *row = i;
                *col = j;
                return;
            }
        }
    }
}
```



```c
        text[i] = toupper(text[i]);

    printf("\nEncrypted Text: ");

    for(i = 0; text[i] != '\0'; i += 2)
    {
        int r1, c1, r2, c2;

        findPosition(text[i], &r1, &c1);
        findPosition(text[i + 1], &r2, &c2);

        if(r1 == r2)
        {
            printf("%c%c", matrix[r1][(c1 + 1) % 5],
                           matrix[r2][(c2 + 1) % 5]);
        }
        else if(c1 == c2)
        {
            printf("%c%c", matrix[(r1 + 1) % 5][c1],
                           matrix[(r2 + 1) % 5][c2]);
        }
        else
        {
            printf("%c%c", matrix[r1][c2],
                           matrix[r2][c1]);
        }
    }

    return 0;
}
```

Output


C:\Users\SAIL\Desktop\Caesar Cipher.exe
Enter plaintext (even number of letters): Hello
Encrypted Text: CFPPMP

Process exited after 23.98 seconds with return value 0
Press any key to continue . . .